

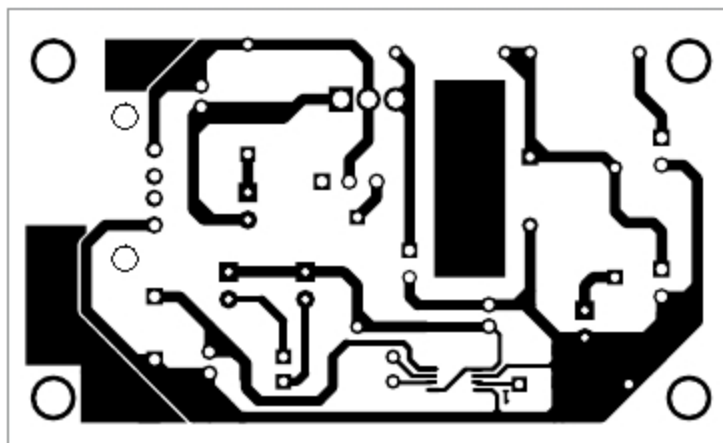


Fig. 3: Actual-size PCB layout of solar USB bicycle power box

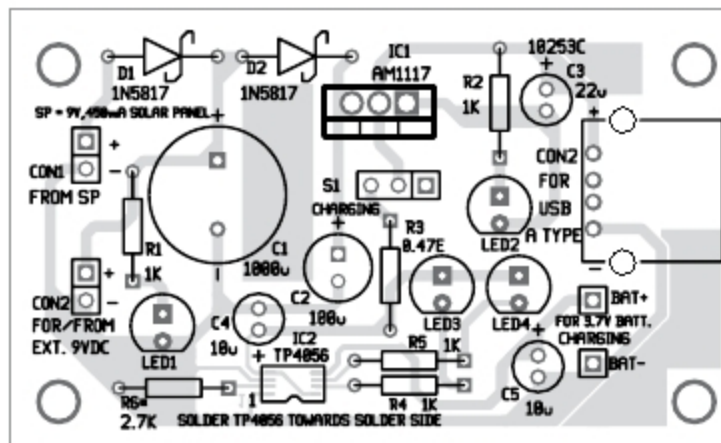


Fig. 4: Components layout for the PCB

Two output terminals (BAT+ and BAT-) extended from the lithium-ion charger circuit allow the charging of the 3.7V lithium-ion spare battery. Here, LED3 lights up when the connected battery is being charged, and LED4 notifies the termination of the battery charging process.

Switch S1 can be used to turn on/off the spare battery charger circuitry. Since a 9V, 450mA solar panel is used, charging current is trimmed at 0.4A using a 2.7-kilo-ohm resistor (R6). However, IC1 and IC2 can handle safe currents up to 1A and, hence, it is

alright to use a higher-current 9V solar panel as an enhancement. In such a case, change the value of R6 to 1.2-kilo-ohm to set a charging current of 1A.

Construction and testing

An actual-size PCB layout for the solar USB bicycle power box circuit is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, connect the 9V solar panel across CON1. (Note that, there are different types of solar panels available in the market that can be suitably mounted on the bicycle for

battery charging.)

Since a surface mount device (SMD)-type component is used here, solder both leadless and leaded components to the board. Pay attention to the polarities of the components, study the data sheets closely and ensure that all components are

placed with specific orientation on the PCB.

After construction and initial testing, fit the entire circuit in an attractive plastic/acrylic enclosure. Prepare the enclosure with holes for connectors, indicators and switches. Place the finished board in the enclosure. Finally, test again after the board has been fully mounted in the enclosure. **EFY**



T.K. Hareendran is founder and promoter of TechNode Protolabz

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